1 **(a)** Smooth sine wave shape passing through $(0, 0)$, $(180, 0)$ and $(360, 0)$ **[B1]**

Maximum turning point at $(90, 1)$ and minimum turning point at $(270, -1)$ **[B1]**

(b) $x = 30$ **[B1]**

$x = 150$ **[B1]**

[Total Marks: 4]

2 **(a)** $\cos x = 0.5$ **[M1]**

$x = 60°$ **[B1]**

$x = 300°$ **[B1]**

(b) 0 (since $\sin x = \dfrac{5}{3} > 1$) **[B1]**

[Total Marks: 4]

3 **(a)** Smooth sine curve passing through $(0, 0), (90°, 1), (180°, 0), (270°, -1), (360°, 0)$ **[B1]**

Smooth cosine curve passing through $(0, 1), (90°, 0), (180°, -1), (270°, 0), (360°, 1)$ **[B1]**

Intersection points shown approximately accurately at $x = 45°$ and $x = 225°$ **[B1]**

(b) $\tan x = 1$ **[M1]**

$x = 45°, 225°$ (Both required) **[A1]**

(c) Identifies the region between the two intersection points **[M1]**

$45° < x < 225°$ **[A1]**

[Total Marks: 7]

4 **(a)** $\sin x = -0.75$ **[M1]**

Reference angle $= 48.6°$ (or $48.59...°$) **[M1]**

$x = 228.6°, 311.4°$ (Both required for the final mark) **[A1]**

(b) x-coordinate is $270°$ **[B1]**

y-coordinate is -1 **[B1]**

[Total Marks: 5]

5 **(a)** $\cos x = -0.25$ **[M1]**

$x = 104.5$ **[A1]**

$x = 360 - 104.47... = 255.5$ **[A1]**

(b) $\tan x = 2.5$ **[M1]**

$x = 68.2$ **[A1]**

$x = 68.19... + 180 = 248.2$ **[A1]**

[Total Marks: 6]

6 **(a)** $\sin 30° = \frac{1}{2}$ leading to $a\left(\frac{1}{2}\right) - b = 0$ or $\frac{b}{a} = \frac{1}{2}$ **[M1]**

$a = 2$ and $b = 1$ **[A1]**

(b) 150 **[B1]**

(c) $k = \cos(242°) = -0.469$ **[B1]**

Uses symmetry of cosine curve: $360° - 242°$ **[M1]**

$x = 118°$ **[A1]**

[Total Marks: 6]

7 $\sin x = \frac{1}{\sqrt{2}}$ **[M1]**

$x = 45°$ **[B1]**

$x = 135°$ **[B1]**

[Total Marks: 3]

8 **(a)** Smooth U-shaped cosine curve starting at $(0, 1)$ and ending at $(4, 1)$ representing 360° **[B1]**

Curve passes through $(1, 0)$, $(3, 0)$ and has a minimum at $(2, -1)$ **[B1]**

(b) $(180°, -1)$ **[B1]**

(c) $\cos x = -0.5$ (or $-\frac{1}{2}$) **[M1]**

$x = 120°$ **[B1]**

$x = 240°$ **[B1]**

[Total Marks: 6]

9 **(a)** $\tan x = -2.8$ **[M1]**

$x = 109.7°$ (accept 109.6° to 109.7°) **[B1]**

$x = 289.7°$ (accept 289.6° to 289.7°) **[B1]**

(b) Factorises to $\sin x(2\sin x - 1) = 0$ **[M1]**

Identifies $\sin x = 0$ or $\sin x = 0.5$ (or $\frac{1}{2}$) **[M1]**

$x = 0°, 180°, 360°$ (all three correct for the mark) **[A1]**

$x = 30°, 150°$ (both correct for the mark) **[A1]**

[Total Marks: 7]